

MIT OpenCourseWare

8.07

Electromagnetism II 8.07 — Electromagnetism II May 2026 **EXTREMELY HARD –****MIT LEVEL****Time Limit: 3 hours****Total Marks: 100**

1. (2 points) Construct an original proof-level problem involving Radiation for 8.07 (Electromagnetism II) and provide a complete solution. Your problem should be at the level of a graduate qualifying exam.
2. (4 points) Analyze the limitations of standard approaches to Relativistic Electrodynamics when applied to edge cases in 8.07. Construct explicit counterexamples showing where intuitive reasoning fails.
3. (2 points) Construct an original proof-level problem involving Radiation Reaction for 8.07 (Electromagnetism II) and provide a complete solution. Your problem should be at the level of a graduate qualifying exam.
4. (4 points) Prove the fundamental theorem concerning Maxwell Stress Tensor in the context of 8.07 (Electromagnetism II). State the theorem precisely, provide a complete proof, and discuss three important corollaries.
5. (2 points) Analyze the limitations of standard approaches to Waveguides when applied to edge cases in 8.07. Construct explicit counterexamples showing where intuitive reasoning fails.
6. (5 points) Analyze the limitations of standard approaches to Radiation when applied to edge cases in 8.07. Construct explicit counterexamples showing where intuitive reasoning fails.
7. (4 points) Prove the fundamental theorem concerning Relativistic Electrodynamics in the context of 8.07 (Electromagnetism II). State the theorem precisely, provide a complete proof, and discuss three important corollaries.
8. (4 points) Construct an original proof-level problem involving Radiation Reaction for 8.07 (Electromagnetism II) and provide a complete solution. Your problem should be at the level of a graduate qualifying exam.

9. (3 points) Analyze the limitations of standard approaches to Maxwell Stress Tensor when applied to edge cases in 8.07. Construct explicit counterexamples showing where intuitive reasoning fails.
10. (3 points) Analyze the limitations of standard approaches to Waveguides when applied to edge cases in 8.07. Construct explicit counterexamples showing where intuitive reasoning fails.
11. (4 points) Analyze the limitations of standard approaches to Radiation when applied to edge cases in 8.07. Construct explicit counterexamples showing where intuitive reasoning fails.
12. (5 points) Analyze the limitations of standard approaches to Relativistic Electrodynamics when applied to edge cases in 8.07. Construct explicit counterexamples showing where intuitive reasoning fails.
13. (5 points) Construct an original proof-level problem involving Radiation Reaction for 8.07 (Electromagnetism II) and provide a complete solution. Your problem should be at the level of a graduate qualifying exam.
14. (4 points) Construct an original proof-level problem involving Maxwell Stress Tensor for 8.07 (Electromagnetism II) and provide a complete solution. Your problem should be at the level of a graduate qualifying exam.
15. (3 points) Construct an original proof-level problem involving Waveguides for 8.07 (Electromagnetism II) and provide a complete solution. Your problem should be at the level of a graduate qualifying exam.
16. (2 points) Construct an original proof-level problem involving Radiation for 8.07 (Electromagnetism II) and provide a complete solution. Your problem should be at the level of a graduate qualifying exam.
17. (6 points) Prove the fundamental theorem concerning Relativistic Electrodynamics in the context of 8.07 (Electromagnetism II). State the theorem precisely, provide a complete proof, and discuss three important corollaries.
18. (6 points) Prove the fundamental theorem concerning Radiation Reaction in the context of 8.07 (Electromagnetism II). State the theorem precisely, provide a complete proof, and discuss three important corollaries.
19. (7 points) Construct an original proof-level problem involving Maxwell Stress Tensor for 8.07 (Electromagnetism II) and provide a complete solution. Your problem should be at the level of a graduate qualifying exam.

20. (3 points) Construct an original proof-level problem involving Waveguides for 8.07 (Electromagnetism II) and provide a complete solution. Your problem should be at the level of a graduate qualifying exam.
21. (4 points) Construct an original proof-level problem involving Radiation for 8.07 (Electromagnetism II) and provide a complete solution. Your problem should be at the level of a graduate qualifying exam.
22. (5 points) Construct an original proof-level problem involving Relativistic Electrodynamics for 8.07 (Electromagnetism II) and provide a complete solution. Your problem should be at the level of a graduate qualifying exam.
23. (3 points) Analyze the limitations of standard approaches to Radiation Reaction when applied to edge cases in 8.07. Construct explicit counterexamples showing where intuitive reasoning fails.
24. (7 points) Analyze the limitations of standard approaches to Maxwell Stress Tensor when applied to edge cases in 8.07. Construct explicit counterexamples showing where intuitive reasoning fails.
25. (3 points) Analyze the limitations of standard approaches to Waveguides when applied to edge cases in 8.07. Construct explicit counterexamples showing where intuitive reasoning fails.

Solutions

Solution 1:

Solution approach: First recall the definitions and key theorems related to Radiation from 8.07. Then apply the appropriate techniques to construct a rigorous argument. The solution must be self-contained and reference only the material from the course.

Solution 2:

The edge case analysis reveals the subtle assumptions underlying standard treatments of Relativistic Electrodynamics. By constructing explicit counterexamples, we see that the usual hypotheses cannot be weakened without loss of the theorem's conclusion.

Solution 3:

Solution approach: First recall the definitions and key theorems related to Radiation Reaction from 8.07. Then apply the appropriate techniques to construct a rigorous argument. The solution must be self-contained and reference only the material from the course.

Solution 4:

The proof follows from the definitions and properties established in 8.07. The key insight is to apply the relevant theorem and verify the hypotheses hold. Detailed solution depends on the specific formulation of the theorem.

Solution 5:

The edge case analysis reveals the subtle assumptions underlying standard treatments of Waveguides. By constructing explicit counterexamples, we see that the usual hypotheses cannot be weakened without loss of the theorem's conclusion.

Solution 6:

The edge case analysis reveals the subtle assumptions underlying standard treatments of Radiation. By constructing explicit counterexamples, we see that the usual hypotheses cannot be weakened without loss of the theorem's conclusion.

Solution 7:

The proof follows from the definitions and properties established in 8.07. The key insight is to apply the relevant theorem and verify the hypotheses hold. Detailed solution depends on the specific formulation of the theorem.

Solution 8:

Solution approach: First recall the definitions and key theorems related to Radiation Reaction from 8.07. Then apply the appropriate techniques to construct a rigorous argument. The solution must be self-contained and reference only the material from the course.

Solution 9:

The edge case analysis reveals the subtle assumptions underlying standard treatments of Maxwell Stress Tensor. By constructing explicit counterexamples, we see that the usual hypotheses cannot be weakened without loss of the theorem's conclusion.

Solution 10:

The edge case analysis reveals the subtle assumptions underlying standard treatments of Waveguides. By constructing explicit counterexamples, we see that the usual hypotheses cannot be weakened without loss of the theorem's conclusion.

Solution 11:

The edge case analysis reveals the subtle assumptions underlying standard treatments of Radiation. By constructing explicit counterexamples, we see that the usual hypotheses cannot be weakened without loss of the theorem's conclusion.

Solution 12:

The edge case analysis reveals the subtle assumptions underlying standard treatments of Relativistic Electrodynamics. By constructing explicit counterexamples, we see that the usual hypotheses cannot be weakened without loss of the theorem's conclusion.

Solution 13:

Solution approach: First recall the definitions and key theorems related to Radiation Reaction from 8.07. Then apply the appropriate techniques to construct a rigorous argument. The solution must be self-contained and reference only the material from the course.

Solution 14:

Solution approach: First recall the definitions and key theorems related to Maxwell Stress Tensor from 8.07. Then apply the appropriate techniques to construct a rigorous argument. The solution must be self-contained and reference only the material from the course.

Solution 15:

Solution approach: First recall the definitions and key theorems related to Waveguides from 8.07. Then apply the appropriate techniques to construct a rigorous argument. The solution must be self-contained and reference only the material from the course.

Solution 16:

Solution approach: First recall the definitions and key theorems related to Radiation from 8.07. Then apply the appropriate techniques to construct a rigorous argument. The solution must be self-contained and reference only the material from the course.

Solution 17:

The proof follows from the definitions and properties established in 8.07. The key insight is to apply the relevant theorem and verify the hypotheses hold. Detailed solution depends on the specific formulation of the theorem.

Solution 18:

The proof follows from the definitions and properties established in 8.07. The key insight is to apply the relevant theorem and verify the hypotheses hold. Detailed solution depends on the specific formulation of the theorem.

Solution 19:

Solution approach: First recall the definitions and key theorems related to Maxwell Stress Tensor from 8.07. Then apply the appropriate techniques to construct a rigorous argument. The solution must be self-contained and reference only the material from the course.

Solution 20:

Solution approach: First recall the definitions and key theorems related to Waveguides from 8.07. Then apply the appropriate techniques to construct a rigorous argument. The solution must be self-contained and reference only the material from the course.

Solution 21:

Solution approach: First recall the definitions and key theorems related to Radiation from 8.07. Then apply the appropriate techniques to construct a rigorous argument. The solution must be self-contained and reference only the material from the course.

Solution 22:

Solution approach: First recall the definitions and key theorems related to Relativistic Electrodynamics from 8.07. Then apply the appropriate techniques to construct a rigorous argument. The solution must be self-contained and reference only the material from the course.

Solution 23:

The edge case analysis reveals the subtle assumptions underlying standard treatments of Radiation Reaction. By constructing explicit counterexamples, we see that the usual hypotheses cannot be weakened without loss of the theorem's conclusion.

Solution 24:

The edge case analysis reveals the subtle assumptions underlying standard treatments of Maxwell Stress Tensor. By constructing explicit counterexamples, we see that the usual hypotheses cannot be weakened without loss of the theorem's conclusion.

Solution 25:

The edge case analysis reveals the subtle assumptions underlying standard treatments of Waveguides. By constructing explicit counterexamples, we see that the usual hypotheses cannot be weakened without loss of the theorem's conclusion.